JOHNSON MUGARRA

Data Scientist | AI/ML Specialist | Banking & Financial Analytics

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with 6+ years of experience designing, deploying, and scaling AI/ML solutions, predictive analytics systems, and automated data pipelines across diverse sectors. Proven expertise in championing AI adoption, developing production-grade models for forecasting, risk assessment, and fraud detection, and translating complex analytics into actionable business insights. Strong background in end-to-end ML lifecycle management (MLOps), building interactive BI dashboards, and leading cross-functional teams to deliver measurable business value. Recognized for excellence in stakeholder engagement, technical training, and driving data-driven decision-making at organizational and policy levels. Ready to leverage transferable analytical expertise to advance Housing Finance Bank's AI strategy and digital transformation.

KEY ACHIEVEMENTS

- Reduced forecast error by 18% through advanced time-series models, demonstrating capability for credit risk and financial forecasting applications
- Compressed evaluation cycles from 6 weeks to under 10 days through ML automation – directly transferable to accelerating banking analytics and reporting
- Achieved 95%+ data reliability in high-stakes evaluations through robust data governance and quality frameworks applicable to banking compliance
- Led technical capacity building workshops on AI/ML, statistical modeling, and BI tools – championing data literacy and AI adoption across diverse teams

CORE TECHNICAL COMPETENCIES

Programming & AI/ML	Python, R, SQL, Java, TensorFlow, PyTorch, Scikit-Learn, Statsmodels, Prophet
ML & Predictive Analytics	Supervised/unsupervised learning, time-series forecasting (applicable to credit scoring & risk modeling), classification models (fraud detection patterns), regression analysis, NLP (sentiment analysis for customer insights), cluster analysis (customer segmentation), propensity modeling
Data Visualization & BI	Power BI (Microsoft Certified), Tableau (Certified Specialist), Looker Studio – expert in executive dashboards, KPI tracking, and data storytelling
Data Engineering & MLOps	ETL pipelines, Apache Spark, Hadoop, Airflow, Docker, Git, API integration, automated model deployment, model monitoring & retraining, PostgreSQL, data warehouse consolidation
Project Management	Agile/Waterfall methodologies, stakeholder management, team leadership, budget control, risk management

PROFESSIONAL EXPERIENCE

Research & MEAL Specialist | Center for Research & Communication (CRC)

Jan 2025 – Present

Leading advanced analytics, AI adoption, and M&E frameworks for data-driven program optimization

- Champion AI and data analytics adoption across multi-sector programs, conducting technical workshops and training sessions on ML/AI, statistical modeling, Python, R, and Power BI – directly aligned with promoting AI awareness and trust
- Supervise cross-functional research teams through end-to-end analytical workflows, improving program effectiveness by 20-30% through data-driven insights and evidence-based decision-making
- Establish robust data governance frameworks, quality assurance protocols, and audit compliance systems – ensuring data security and regulatory compliance across organizational partnerships
- Collaborate with diverse stakeholders to align data strategies with organizational objectives, promoting knowledge sharing and adaptive management – skills directly transferable to banking operations

Data Scientist | MAAIF, VANEX, Makerere University & CRS

Jan 2024 – May 2025

Led ML model development, deployment, and MLOps for production-grade predictive analytics systems

- Designed and deployed end-to-end predictive modeling system integrating multiple data sources (API, satellite, ground data), achieving 18% MAE reduction and >80% accuracy – demonstrating capabilities for credit scoring, risk assessment, and financial forecasting in banking
- Led complete ML lifecycle from data collection through model development, validation, deployment, and monitoring – implementing automated retraining protocols and ensuring ethical AI compliance and scalability
- Built and automated scalable ETL pipelines with daily API synchronization using Python + Airflow, ensuring secure data consolidation in centralized data warehouse – ready for banking core system integration
- Developed interactive Tableau/Power BI dashboards enabling non-technical stakeholders to access real-time insights and make data-driven decisions – applicable to banking executive reporting and business intelligence
- Presented findings and policy recommendations to H.E. President Yoweri Kaguta Museveni (April 2025) – demonstrating exceptional communication skills and ability to translate complex analytics for executive leadership

Research Associate | USAID-SITES, SSS Inc. & DLH Corporation

Jun 2023 – May 2025

Delivering advanced analytics, automated reporting, and impact evaluation for high-stakes programs

- Supervised digital data collection and performed rigorous SQL/Python data quality audits, achieving 95%+ data reliability for critical impact evaluations – transferable to banking data integrity and compliance
- Developed stakeholder-focused interactive dashboards and automated reports that enabled 15% operational efficiency gains – demonstrating ability to leverage ML for business process optimization
- Built robust ETL pipelines between multiple systems (SurveyCTO, Power BI, PostgreSQL) with daily API automation – ready for integration with core banking systems, CRM, and digital channels

Data Analyst / Research Associate | Bronkar (U) Limited

Jul 2021 – Jun 2023

Mixed-methods analytics, M&E frameworks, and data governance across East Africa

- Designed comprehensive M&E frameworks and integrated digital data collection tools, reducing field operations time by 25% – demonstrating efficiency optimization through analytics
- Conducted advanced statistical analysis and produced evidence-based recommendations that improved outcomes for stakeholders – showcasing analytical rigor and business impact focus
- Led technical capacity building and staff training on analytics tools, building sustainable organizational data capabilities – aligned with championing AI adoption

SELECTED AI/ML PROJECTS (Banking Applications)

Predictive Analytics & Forecasting System *(2024-2025)*

- Engineered full-stack time-series forecasting pipeline with multi-source data integration, achieving 18% MAE improvement – directly applicable to credit risk modeling, loan default prediction, and financial forecasting
- Implemented Prophet with hierarchical reconciliation for scalable predictions – techniques transferable to customer lifetime value estimation and churn prediction in banking
- Deployed interactive real-time dashboard for stakeholder decision-making – similar to executive banking dashboards for portfolio monitoring and risk management

Automated ML Pipeline for Impact Evaluation *(2023-2026)*

- Built Python + Airflow ML pipeline with automated data ingestion, causal inference modeling, and auto-generated reports – reducing cycle time by 85% (from 6 weeks to <10 days)
- Demonstrates capability to automate banking reports for business process optimization, fraud pattern detection, and operational analytics at scale
- Implemented robust data quality checks and daily batch API automation – ensuring data accuracy for regulatory compliance and risk management

NLP & Customer Insights Analytics

- Applied NLP and sentiment analysis to conversational datasets for evidence-based decision-making – applicable to analyzing customer feedback, call center data, and digital channel interactions in banking

EDUCATION

Master of Science in Data Science

Manipal Academy of Higher Education (MAHE), Karnataka, India

Sep 2022 – Oct 2024

Relevant Coursework: Machine Learning, Predictive Analytics, Time-Series Forecasting, NLP, Big Data Analytics, Advanced Regression, Statistical Modeling, Data Visualization

Bachelor of Science in Agricultural Science

Makerere University, Kampala, Uganda

Sep 2016 – May 2021

Relevant Coursework: Biometrics & Statistics, Research Methods, Mathematics

PROFESSIONAL CERTIFICATIONS

- Google Certified: Project Management Professional (PMP)
- Microsoft Certified: Power BI Data Analyst Associate
- SAS Certified Statistical Business Analyst
- Tableau Desktop Specialist

LEADERSHIP & BUSINESS COMPETENCIES

- AI Strategy & Adoption: Champion AI awareness, trust, and implementation across organizational departments through training, workshops, and knowledge-sharing sessions
- Team Leadership: Supervised cross-functional teams, managed performance, and built technical capabilities across diverse analytical projects
- Stakeholder Engagement: Proven track record presenting complex analytics to executive leadership, government officials, and international partners – translating technical insights into business value
- Budget & Cost Control: Experience managing project budgets and controlling costs within allocated resources
- Communication & Conflict Resolution: Exceptional interpersonal skills with collaborative, open approach to working in environments where ideas are shared and challenged
- Ethical AI & Compliance: Strong understanding of ethical AI implementation, data governance, regulatory compliance, and risk management

AWARDS & RECOGNITION

- **Presidential Presentation (April 2025):** Selected to present advanced analytics research findings and policy recommendations to H.E. President Yoweri Kaguta Museveni at Makerere University – reflecting national-level recognition of analytical excellence and business impact

VALUE PROPOSITION FOR HOUSING FINANCE BANK

I bring proven expertise in developing production-grade ML models, and translating complex analytics into actionable business insights. My experience spans the complete ML lifecycle – from identifying high-value use cases through deployment, monitoring, and continuous improvement. I have demonstrated success in:

- Leading AI strategy and driving organizational adoption through training and knowledge-sharing
- Building predictive models with direct applications to credit scoring, fraud detection, churn prediction, and customer lifetime value
- Developing interactive dashboards that enable rapid, data-driven decision-making
- Implementing robust data governance, quality assurance, and secure data warehouse consolidation
- Automating reports and optimizing business processes through ML – delivering measurable efficiency gains

I am eager to leverage my transferable analytical expertise to advance Housing Finance Bank's AI strategy, bridge the gap between advanced analytics and business value, and deliver measurable impact aligned with your strategic goals.